

PROJECTS (for more updates and complete projects structure check github repositories by clicking on this [GitHubLink](#))

Fitness Microservices App (currently working)

- Built a full-stack, microservices-based fitness platform with six services (Eureka, Config Server, API Gateway, User, Activity, AI) enabling activity tracking, RESTful APIs, AI-driven recommendations (Google Gemini), and a React + Vite frontend with Redux and MUI.
- Addressed service discovery, config complexity, auth, and async processing by implementing Eureka, centralized Config Server, Keycloak OAuth2, and RabbitMQ; used PostgreSQL + MongoDB to balance relational and document storage.
- Future Development Plans: Roadmap includes Redis caching, Kubernetes orchestration, CI/CD pipelines, advanced AI/ML analytics, mobile apps with real-time tracking, and enhanced security/GDPR compliance.
- Tools and Technology used - Java (Java 26), Spring Boot, Spring Cloud (Eureka, Config Server, API Gateway), Keycloak, RabbitMQ, PostgreSQL, MongoDB, Google Gemini API, React (Vite, Redux, MUI), Docker, AWS, Maven, Lombok, ESLint.

Hand Sign Language Recognition System

- Developed a Hand Sign Language Recognition System using Convolutional Neural Networks (CNNs).
- Captured raw hand images through a Hand Tracking model for preprocessing
- Converted images to black-and-white format to isolate hand boundaries from the background also Implemented alternate method where all the joints in hands are tracked
- Implemented classification algorithms to map hand sign symbols to their corresponding meanings
- Currently enhancing the system by integrating multiple models for the same input and combining outputs to improve classification accuracy (also working on optimising the model for system with less resources)
- Tools and Technologies - Jupyter Notebook, Python, CNN, Image Processing, Data Pre-processing, OpenCV, TensorFlow, Keras

Thyroid Detection System

- Implemented and compared multiple ML models (Regression, Logistic Regression, KNN, SVM, Decision Tree, Random Forest, Naive Bayes, DBSCAN, XGBoost, ANN) for thyroid disease detection
- Applied advanced data preprocessing and filtration techniques, selected the best performing model based on F1-Score and Accuracy to ensure reliable predictions
- Tools and Technology used - Jupyter Notebook, Python, Numpy, Pandas, Data Visualisation, Scikit-learn, Matplotlib, TensorFlow

TECHNICAL SKILLS and INTERESTS

Languages: Java, Python, SQL

Framework Libraries: SpringBoot, MicroServices, Docker, Kubernetes, GitHub, Pandas, Numpy, Tensorflow, Keras, Matplotlib, Hugging Face, OpenCV, Matplotlib, Torch, RESTful API

Tools: Jupyter Notebook, VS Code, Anaconda, IntelliJ IDEA

Cloud/Databases: SQL, PostgreSQL, AWS

Relevant Coursework: Machine Learning, Natural Language Processing, Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Computer Networks, Software Engineering

Areas of Interest: Software Development, Machine Learning and AI

Soft Skills: Leadership, Problem Solving, Self-learning, Presentation, Adaptability, Teamwork

COURSES and CERTIFICATIONS

Deep Learning (IIT Ropar, NPTEL, 2025)

Generative AI (Syllogistek Systems Private Limited, 2024)

Data Science, ML & Deep Learning Using Python (Syllogistek Systems Private Limited, 2023)

Full Stack PHP (Syllogistek Systems Private Limited, 2022)

POSITION OF RESPONSIBILITY

Joint Secretary of University Student Council (Awarded Best Council Member 2024)

Member of Green Club (Organised Multiple Planting Drives)

Member of Blood Donation Club

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2021-2025

Silicon Institute of Technology, Silicon University | Bhubaneswar, Odisha, India

CGPA: 7.97

Intermediate (Class XII)

2020-2021

Sankalp Academy | Jhalawar, Rajasthan, India

Percentage: 73

Matriculate (Class X)

2017-2018

Lady Anusuiya Singhanian Education Center | Jhalawar, Rajasthan, India

Percentage: 80